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ORIGINAL ARTICLE

Annual Report to the Nation on the Status of Cancer, part 2: Early assessment of the COVID-19 pandemic's impact on cancer diagnosis

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Abstract

Background: With access to cancer care services limited because of coronavirus disease 2019 control measures, cancer diagnosis and treatment have been delayed. The authors explored changes in the counts of US incident cases by cancer type, age, sex, race, and disease stage in 2020.

Methods: Data were extracted from selected US population-based cancer registries for diagnosis years 2015–2020 using first-submission data from the North American Association of Central Cancer Registries. After a quality assessment, the monthly numbers of newly diagnosed cancer cases were extracted for six cancer types: colorectal, female breast, lung, pancreas, prostate, and thyroid. The observed numbers of incident cancer cases in 2020 were compared with the estimated numbers by calculating observed-to-expected (O/E) ratios. The expected numbers of incident cases were extrapolated using Joinpoint trend models.

Results: The authors report an O/E ratio <1.0 for major screening-eligible cancer sites, indicating fewer newly diagnosed cases than expected in 2020. The O/E ratios were lowest in April 2020. For every cancer site except pancreas, Asians/Pacific Islanders had the lowest O/E ratio of any race group. O/E ratios were lower for cases diagnosed at localized stages than for cases diagnosed at advanced stages.

Conclusions: The current analysis provides strong evidence for declines in cancer diagnoses, relative to the expected numbers, between March and May of 2020. The declines correlate with reductions in pathology reports and are greater for cases diagnosed at in situ and localized stage, triggering concerns about potential poor cancer outcomes in the coming years, especially in Asians/Pacific Islanders.

Plain Language Summary

- To help control the spread of coronavirus disease 2019 (COVID-19), health care organizations suspended nonessential medical procedures, including preventive cancer screening, during early 2020.
- Many individuals canceled or postponed cancer screening, potentially delaying cancer diagnosis.
- This study examines the impact of the COVID-19 pandemic on the number of newly diagnosed cancer cases in 2020 using first-submission, population-based cancer registry database.
- The monthly numbers of newly diagnosed cancer cases in 2020 were compared with the expected numbers based on past trends for six cancer sites.
- April 2020 had the sharpest decrease in cases compared with previous years, most likely because of the COVID-19 pandemic.

KEYWORDS
cancer incidence, cancer stage, cancer surveillance, coronavirus disease 2019 (COVID-19), observed-to-expected, pathology reports

INTRODUCTION

Access to timely cancer care continues to be a concern as the coronavirus disease 2019 (COVID-19) pandemic lingers worldwide. In early 2020, several health professional organizations and governmental agencies issued guidance suggesting that all elective surgeries and nonessential medical procedures, including cancer screening, be suspended because of the severe acute respiratory syndrome coronavirus 2 (SARS–CoV-2) outbreak.[1] Evidence documented that the use of cancer screenings across the United States declined at the beginning of the COVID-19 pandemic.[2] Although screening use has since increased, it has not reached prepandemic expected levels.[3] Suspension of cancer-related procedures has created a backlog of health services, such as increasing wait times for cancer surgery.[4,5] Despite the Biden administration's plan to declare the public health emergency over on May 11, 2023,[6] it might be some time before cancer services return to prepandemic volumes.

Population-based cancer registries are in a unique position to detect changes in cancer trends and patterns that could be related to COVID-19 pandemic control interventions. Changes in the number and stage distribution of cancer cases reported to registries could be an informative first indication of the COVID-19 pandemic's impact on the cancer care continuum. This study analyzes cancer case counts and pathology reports submitted to central cancer registries for diagnosis year 2020 and compares these counts with expected counts based on the past trends to understand whether early evidence points toward changes in cancer incidence during the early phase of the COVID-19 pandemic.

MATERIALS AND METHODS

Registry and case selection

Data about newly diagnosed cancer cases were obtained from the North American Association of Central Cancer Registries (NAACCR) database, which comprises data from population-based registries that participate in the Centers for Disease Control and Prevention's National Program of Cancer Registries and/or the National Cancer Institute's (NCI's) Surveillance, Epidemiology, and End Results (SEER) Program. Case counts for 2015–2020 were extracted from the NAACCR database using SEER*Stat software (8.4.0.1). Registries report newly diagnosed cases first at 12 months after the end of the diagnosis year, and then with every successive annual call for data. Therefore, cases diagnosed in 2015 were first reported to NAACCR at the end of 2016 and then resubmitted at the end of 2017. Similarly, cases diagnosed during 2020 were first submitted at the end of 2021 and were resubmitted in 2022 and later. Registries with a case count ratio >0.7 for a specific cancer site between the first and second submission for five successive submissions (years 2015 through 2019) were considered high quality for 12-month case capture and were included in the analysis for that cancer site. We conducted an assessment of the above-mentioned ratio for the selected NAACCR registries for the following six primary sites: colorectal (*International Classification of Diseases for Oncology*, third edition codes C180, C182–C189, C199, and C209),[7] female breast (C500–C506 and C508–C509),[7] lung (C340–C343 and C348–C349),[7] pancreas (C250–C254 and C257–C259),[7] prostate (C619),[7] and thyroid (C739).[7] The selection of sites included cancers with screening recommendations (female

TABLE 1 Population-based cancer registries with high 12-month data completeness by cancer site, age, and sex categories of interest; percent coverage of the US population (2020); and average annual case count (2015–2019).

Cancer site	Sex/age group	Registries included, state abbreviation	US population covered, 2020 (%)[a]	Average annual case count, 2015–2019[b]
Colorectal	Male and female/ 50–74 years	Alabama, Georgia, Iowa, Kentucky, Louisiana, Maine, Montana, Nebraska, New York, North Carolina, North Dakota, Ohio, Oklahoma, West Virginia, Wyoming	25	3649
Breast	Female/40–74 years	Alabama, Delaware, Georgia, Iowa, Kentucky, Louisiana, Maine, Mississippi, Montana, Nebraska, New York, North Carolina, North Dakota, Ohio, Oklahoma, West Virginia, Wyoming	26	11,139
Lung	Male and female/ 55–74 years	Kentucky, Louisiana, Maine, Mississippi, Montana, New York, North Carolina, North Dakota, Ohio, Oklahoma, West Virginia	20	5106
Pancreas	Male and female/ 50–74 years	Kentucky, Louisiana, Maine, Mississippi, Montana, New York, North Carolina, North Dakota, Ohio, Oklahoma, West Virginia	20	1214
Prostate	Male/50–74 years	Georgia, Kentucky, Louisiana, Maine, Montana, Nebraska, New York, North Carolina, North Dakota, West Virginia, Wyoming	18	6037
Thyroid	Male and female/ 50–74 years	Alabama, Delaware, Georgia, Kentucky, Louisiana, Maine, Mississippi, Montana, New York, North Carolina, North Dakota, Ohio, Oklahoma, West Virginia, Wyoming	25	971

[a]Calculated using the Surveillance, Epidemiology, and End Results (SEER) Program (www.seer.cancer.gov) SEER*Stat Database: Populations—Total United States (1969–2020) <Katrina/Rita Adjustment>—Linked To County Attributes-Total United States, 1969–2020 Counties. National Cancer Institute, Division of Cancer Control and Population Sciences, Surveillance Research Program, released January 2022.

[b]Calculated using the SEER*Stat Database: North American Association of Central Cancer Registries Incidence Data—Certification year and 12-month data, 2014–2020, for the United States (which includes data from the Centers for Disease Control and Prevention's National Program of Cancer Registries, the Canadian Cancer Registry's Provincial and Territorial Registries, and the National Cancer Institute's SEER Program registries), certified by the North American Association of Central Cancer Registries as meeting high-quality incidence data standards for the specified time periods; submitted December 2021.

breast, lung, colorectal), cancers with frequent incidental detection (prostate, thyroid), and a cancer with rapid progression and unfavorable outcomes (pancreas). Male breast cancer cases were excluded because of small numbers and the absence of a recommendation for screening.

The number of registries selected and the population covered by the selected registries varied by cancer site, as indicated in Table 1. For the purposes of this analysis, age groups were selected based on the American Cancer Society's screening recommendations.[8] To facilitate comparison, similar age groups were used for cancer sites without a screening recommendation (Table 1). The highest population coverage was observed for female breast cancer (17 registries covering 26% of the US population in the age category of interest [ages 40–74 years]), with an average annual case count of 11,139. The lowest coverage was observed for prostate cancer (11 registries covering 18% of the US male population ages 50–74 years).

All cancer reportability criteria for cases included in this study followed the NAACCR standards.[9] Similarly, all data elements used for the analysis conformed to the NAACCR Data Dictionary.[10]

Case counts were stratified by NAACCR month and year of diagnosis, sex, race (NAACCR recode), and stage (SEER Summary Stage 2000 for diagnosis years 2015–2017 and Summary Stage 2018 for diagnosis years 2018–2020). Race was categorized into four groups: White, Black, Asian or Pacific Islander (API), and American Indian or Alaska Native (AIAN). Data for Hispanic origin were not available in this data set because they were not required by NAACCR for this particular call for data. Cancer stage was categorized as in situ, localized, advanced (regional and distant), and unknown.

Statistical analysis and modeling

To evaluate the impact of the COVID-19 pandemic on cancer incidence counts, we calculated ratios of the observed-to-expected (O/E) number of cancer diagnoses (O/E ratio) using the NCI's Joinpoint Regression Program, version 4.9.0.0.[11] The default settings were used to quantify temporal trends from 2015 to 2019 based on log-scale with Poisson variance. Data were stratified by month of diagnosis and revealed monthly fluctuations in cases. To account for such variation, we modeled each month separately. The last segment from the model was extrapolated to 2020 and was considered the expected 2020 count in the absence of the COVID-19 pandemic. The sum of the projected counts for all 12 months was considered the expected count for 2020. An O/E ratio close to 1.0 suggests that the 2020 observed count is in line with data from previous years. If the O/E ratio is <1.0, this suggests that there were fewer cases reported than expected. The standard error for the ratio was obtained using the delta method,[12] which was then used to determine the *p* value for testing whether the O/E ratio is different from 1.0 as well as the 95% confidence interval (CI) for each O/E ratio; these 95% CIs also allow for informal comparisons between groups without specifying a referent group.

Pathology report volume comparison

Electronic pathology reports for specimens collected for the period from January 2019 through December 2020 from 11 participating central cancer registries (Connecticut, Georgia, Hawaii, Iowa, Kentucky, Louisiana, New Jersey, New Mexico, New York, Seattle, and Utah) were compiled by the SEER Program for a separate analysis. Central cancer registries participating in the program received pathology reports transmitted electronically in real time; therefore, the acquisition of pathology reports has not been affected by functional delays that could have affected cancer abstraction in hospital registries. To ensure consistency over time, we limited the analysis to facilities that transmitted reports every month between 2018 and 2020. A convolutional neural network algorithm was used to classify reports by the site and histology of the primary tumor (female breast, lung, colorectal, pancreas, prostate, thyroid) and by report type (biopsy, cytology, molecular and biomarker, surgical, blood/bone marrow/flow cytometry, and autopsy).[13] The date of specimen collection was used to subset and categorize the reports by the month of collection. Reportable biopsy counts by month of collection were compared with NAACCR case counts by cancer site and month of diagnosis between January 2019 and December 2020.

RESULTS

Figure 1 presents the O/E ratio for newly diagnosed cases calculated for each cancer by month of diagnosis in 2020. Between April and May, registries recorded significantly fewer cases than expected for every cancer type. For all six cancer types, April 2020 was the month with the lowest ratio. The most dramatic decline in the O/E ratio was seen for cancers that generally have a more favorable prognosis: thyroid, prostate, female breast, and colorectal. In contrast, the decline in O/E ratio was markedly attenuated for cancers with poor survival: pancreas and lung. Although the O/E ratios for most cancer sites clearly declined in March 2020, the O/E ratio for female breast cancer was an exception; it appeared that a longer lead time was necessary for the ratio to change in female breast cancer than in other sites.

Figure 2 shows that the variation in counts of newly diagnosed cases correlated with changes in the counts of biopsy pathology reports transmitted to the registries. Not every case reported to a registry has been confirmed by a biopsy. In addition, a case may have multiple biopsy reports; therefore, there is no predetermined number of pathology reports per case. A dramatic decline in pathology report counts was seen starting in March 2020 for all cancer sites. Some rebound that exceeded 2019 pathology report counts were seen for pancreas, prostate, and thyroid cancers.

When data were aggregated over all 12 months of the year 2020, the observed number of newly diagnosed cases was statistically significantly lower than the expected number for all major cancer sites eligible for screening (colorectal, female breast, lung, and prostate cancers; Figure 3).

The O/E ratio was 0.90 for colorectal cancer (females and males), 0.89 for prostate cancer, 0.92 for female breast cancer, and 0.94 for lung cancer (males and females). Among screening-eligible cancers, the absolute difference in the O/E ratio for males versus females was <0.1, indicating little difference by sex. We calculated the O/E ratio for two cancer sites with no screening recommendations: pancreas and thyroid. Among males, we observed no statistically significant difference in observed versus expected counts for those sites. However, lower than expected counts were recorded for females with thyroid cancer (statistically significant) and pancreatic cancer (not statistically significant).

Figure 4 illustrates the O/E ratios for newly diagnosed cases by cancer site and race category. Among the four race groups, API persons had the lowest O/E ratio for every cancer site except pancreas. The CI for the pancreatic cancer O/E ratio included one for every race category. In contrast, the CI for the lung cancer O/E ratio excluded one for every race group. In fact, significantly fewer cases than expected were observed among White, Black, and API persons for each cancer site included in this analysis, with the exception of pancreas and thyroid among White persons. Although O/E ratios for AIAN persons were generally higher compared with the other race groups, the wide CIs made it difficult to interpret these results.

Figure 5 presents the O/E ratios for newly diagnosed cases by cancer site and age category. For colorectal, breast, and prostate cancers, there were fewer diagnosed cases than expected among people aged 40 years or older, but not among people younger than 40 years. For lung and thyroid cancers, there were fewer diagnosed cases than expected among all age groups. The number of pancreatic cancer cases was as expected, regardless of age. The variation in the O/E ratio between age categories was small for all other cancer sites, except for lung cancer, in which the ratio was particularly low for cases diagnosed in patients younger than 40 years.

Table 2 shows the O/E ratio for newly diagnosed cases by cancer site and stage at diagnosis. A clear pattern emerges, with every cancer site showing a lower O/E ratio for localized cancer than for advanced cancer. This indicates that proportionally fewer localized cases were diagnosed during the year 2020. Cancer sites with no screening recommendations (i.e., thyroid and pancreas) showed no decline from expected in the number of advanced cases, a 6%–10% decline in localized case counts, and a 62% decline in in situ cases for thyroid. Excluding the *unknown* stage category, the most dramatic declines from the expected number of cases (lowest O/E ratios) were seen for in situ and localized cases of female breast (localized only), colorectal, lung, and prostate cancers.

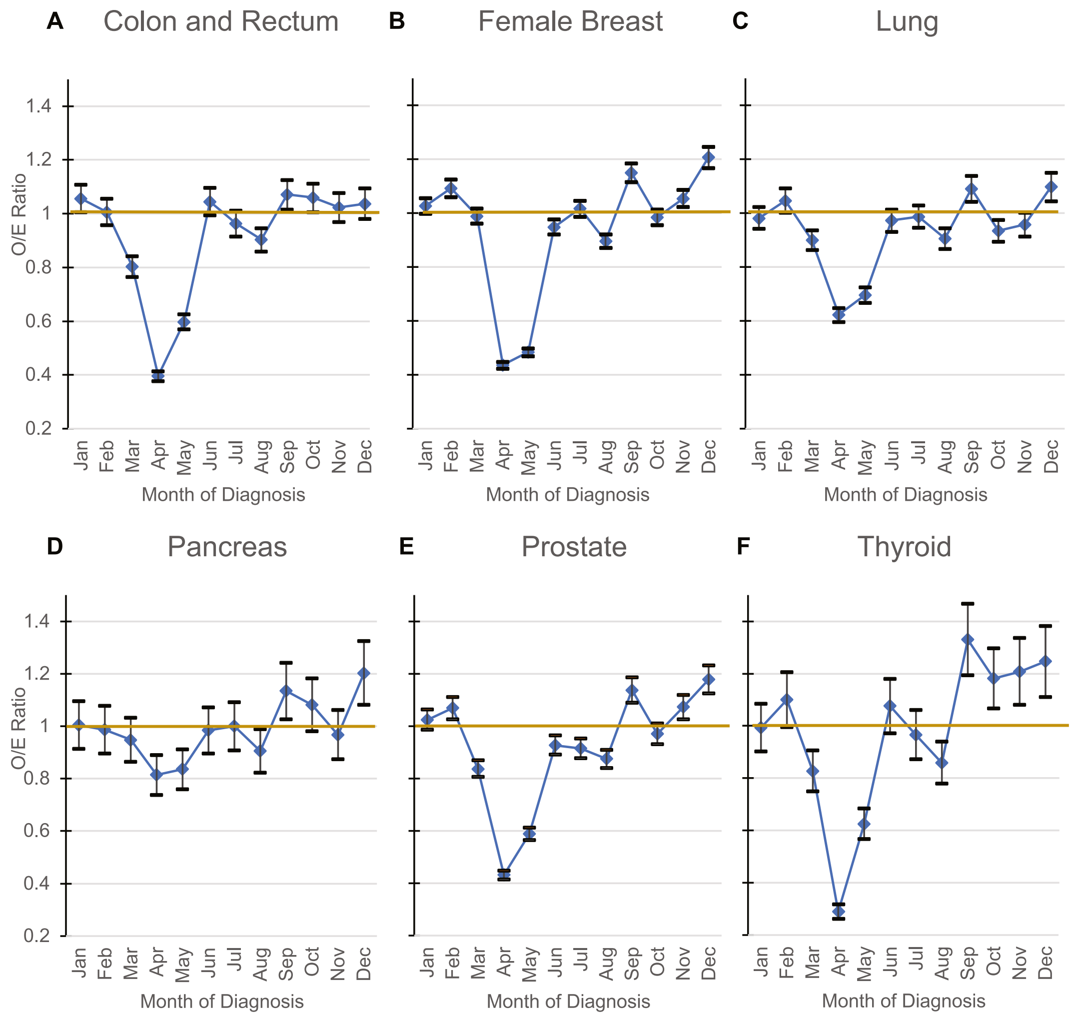

FIGURE 1 (A–F) Observed-to-expected (O/E) ratios for newly diagnosed cancer cases, diagnosis year 2020, by month of diagnosis and cancer site. The observed 2020 count was compared with the expected count by ratio (O/E ratio). The standard error for the ratio was obtained using the delta method to determine the *p* value and 95% confidence interval for each O/E ratio. Data source for observed counts: SEER*Stat Database, North American Association of Central Cancer Registries Incidence Data—Certification year and 12-month data, 2014–2020, for the United States (which includes data from the Centers for Disease Control and Prevention's National Program of Cancer Registries, the Canadian Cancer Registry's Provincial and Territorial Registries, and the National Cancer Institute's SEER Program registries), certified by the North American Association of Central Cancer Registries as meeting high-quality incidence data standards for the specified time periods; submitted December 2021. SEER indicates Surveillance, Epidemiology, and End Results.

DISCUSSION

Decline in cancer case counts and pathology reports

By using population-based cancer registry data, we report that the number of newly diagnosed cases of six major cancers declined during the first calendar year of the COVID-19 pandemic in the United States. Specifically, this investigation quantifies differences between observed versus expected (O/E) cases reported to cancer registries. For most cancer types and population groups analyzed, the incident cases decreased abruptly starting in March 2020, coinciding with the COVID-19 mitigation efforts. The disparity between observed and expected cases was most pronounced in April and improved minimally in May. Subsequently, for most cancer types, the deficit decreased in June and generally was not measurable in the second half of 2020. Furthermore, we report some evidence of a rebound effect for certain cancer types, although the rebound did not offset the deficit in the first one half of the year. This study provides quantitative information on the impact of COVID-19 on the diagnosis of major cancers during the first year of the pandemic.

The decreases in cases paralleled the declines in the numbers of pathology reports. These reports are transmitted to central cancer registries automatically in real time and require little to no additional processing effort from registry staff. Currently, the NCI SEER Program can conduct real-time monitoring of pathology report trends using a suite of tools developed by the NCI in collaboration with the Department of Energy as part of the *Joint Design of Advanced Computing Solutions for Cancer*.[14,15] The transmission mechanisms of pathology reports differ from those of other registry data streams,

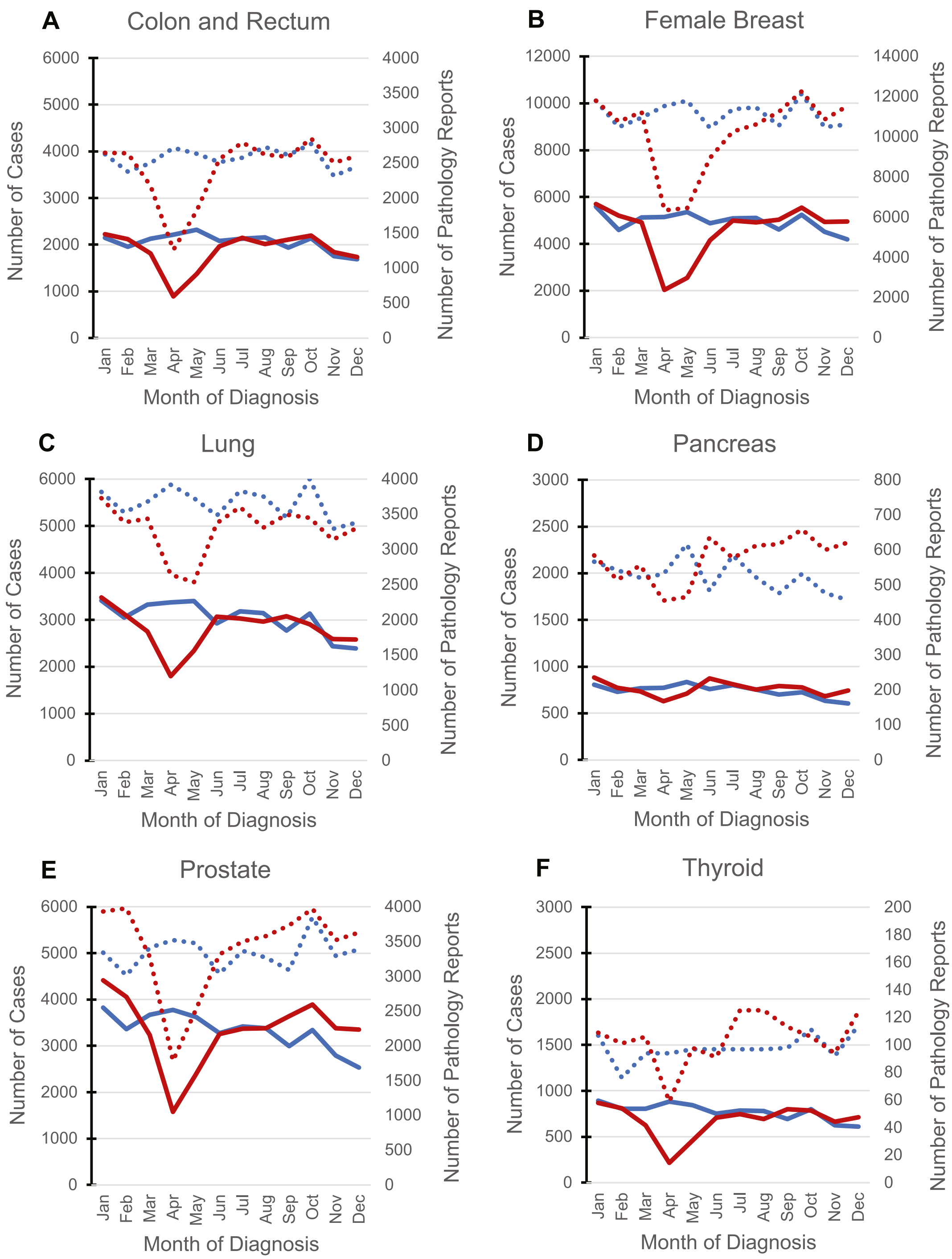

FIGURE 2 (A–F) Monthly number of reportable biopsy pathology reports versus newly diagnosed cancer cases by cancer site in 2019 and 2020. The frequency distribution of cases is compared with the frequency distribution of reportable biopsy pathology reports in 2019 and 2020 by month of diagnosis for six cancer sites. SEER*Stat Database, North American Association of Central Cancer Registries Incidence Data—Certification year and 12-month data, 2014–2020, for the United States (which includes data from the Centers for Disease Control and Prevention's National Program of Cancer Registries, the Canadian Cancer Registry's Provincial and Territorial Registries, and the National Cancer Institute's SEER Program registries), certified by the North American Association of Central Cancer Registries as meeting high-quality incidence data standards for the specified time periods; submitted December 2021. SEER indicates Surveillance, Epidemiology and End Results.

such as the transmission of cancer abstracts, which could have been delayed by operational challenges like staff shortages or limited physical access to medical records. Therefore, the reductions in pathology reports in the first one half of 2020 are most likely related to decreases in screening procedures,[16,17] rescheduling of surgical procedures,[18] and fewer in-person medical visits.[19]

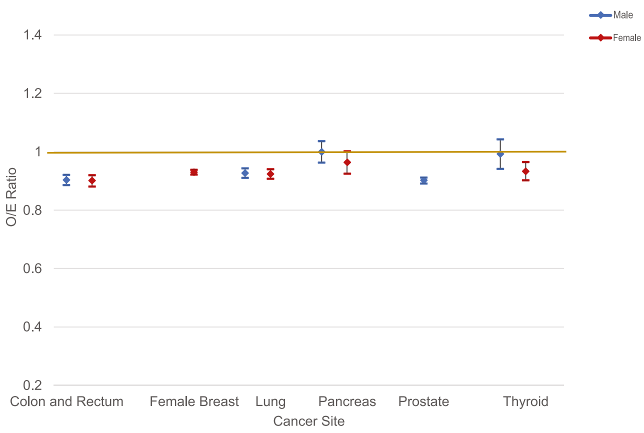

FIGURE 3 Sex-specific observed-to-expected (O/E) ratios for newly diagnosed cancer cases by cancer site in 2020. The observed 2020 count was compared with the expected count by ratio (O/E ratio). The standard error for the ratio was obtained using the delta method to determine the *p* value and 95% confidence interval for each O/E ratio. Data source for observed counts: SEER*Stat Database: North American Association of Central Cancer Registries Incidence Data—Certification year and 12-month data, 2014–2020, for the United States (which includes data from the Centers for Disease Control and Prevention's National Program of Cancer Registries, the Canadian Cancer Registry's Provincial and Territorial Registries, and the National Cancer Institute's SEER Program registries), certified by the North American Association of Central Cancer Registries as meeting high-quality incidence data standards for the specified time periods; submitted December 2021.

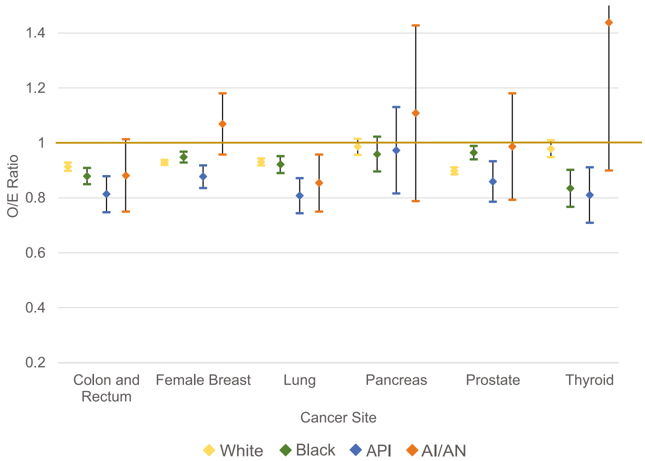

FIGURE 4 Race-specific observed-to-expected (O/E) ratios for newly diagnosed cancer cases by cancer site in 2020. The observed 2020 case count was compared with the expected count by ratio (O/E ratio). The standard error for the ratio was obtained using the delta method to determine the *p* value and 95% confidence interval for each O/E ratio. Data source for observed counts: SEER*Stat Database, North American Association of Central Cancer Registries Incidence Data—Certification year and 12-month data, 2014–2020, for the United States (which includes data from the Centers for Disease Control and Prevention's National Program of Cancer Registries, the Canadian Cancer Registry's Provincial and Territorial Registries, and the National Cancer Institute's SEER Program registries), certified by the North American Association of Central Cancer Registries as meeting high-quality incidence data standards for the specified time periods; submitted December 2021. AIAN indicates American Indian or Alaska Native; API, Asian or Pacific Islander; SEER, Surveillance, Epidemiology, and End Results.

Decline varies across demographic groups

For screen-detected cancers, the O/E ratio was similar for males and females. However, a sex-related association was observed for thyroid and pancreatic cancers. There is evidence that subclinical papillary thyroid tumors are diagnosed more often in females than in males.[20] Subclinical diagnosis is frequently the result of an ultrasound examination. Several articles reported a decline in the use of ultrasound in emergency departments during 2020.[21,22] Other reports asserted that the decrease in the number of imaging examinations was even sharper in ambulatory settings than in hospitals.[23] Pancreatic cancer is often detected because of acute abdominal pain and symptoms of biliary blockage. Although the proportion of cases diagnosed with regional or distant disease is approximately the same in males and females, there is evidence that males have more comorbidities at the time of diagnosis and are diagnosed at younger ages.[24] Given the awareness that patients with multiple comorbidities are at higher risk of severe disease,[25] it is possible that male patients with nonspecific symptoms were more likely to be hospitalized or that they received risk-based priority in outpatient care settings. This may explain in part why more pancreatic cancer cases were reported in males than in females. In contrast, the recommendations to delay screening applied to all patients irrespective of sex; therefore, the declines in newly diagnosed cases of those cancers were observed almost equally in males and females.

In the United States, Black, AIAN, Pacific Islander, and Hispanic and Latino persons have experienced higher rates of infection, hospitalization, and deaths from SARS–CoV-2 than non-Hispanic White persons, whereas Asian persons have experienced lower rates.[26–28] Compared with White persons, the observed cancer case count among API persons tended to be lower than the expected count for colorectal, female breast, lung, and thyroid cancers. Various financial, linguistic, and cultural barriers had prevented Asian immigrants' use of health care services before the COVID-19 pandemic.[29] In general, during the prepandemic era, there was low cancer screening prevalence among API persons, even after the enactment of the Patient Protection and Affordable Care Act, which required most insurance plans to provide certain recommended cancer screenings at no cost. This was exacerbated by the COVID-19 pandemic, with API persons having the largest decline in cancer screening compared with other racial/ethnic groups, and no rebound in sight.[30,31] Several factors may have contributed to this decline, in particular the psychological impact of anti-Asian stigma because of the SARS–CoV-2 outbreak.[32] This could explain why fewer than expected cases were diagnosed in the Asian community for cancers diagnosed through screening. In contrast, Black persons experienced a smaller decline in observed prostate cancer cases than White persons. The less pronounced

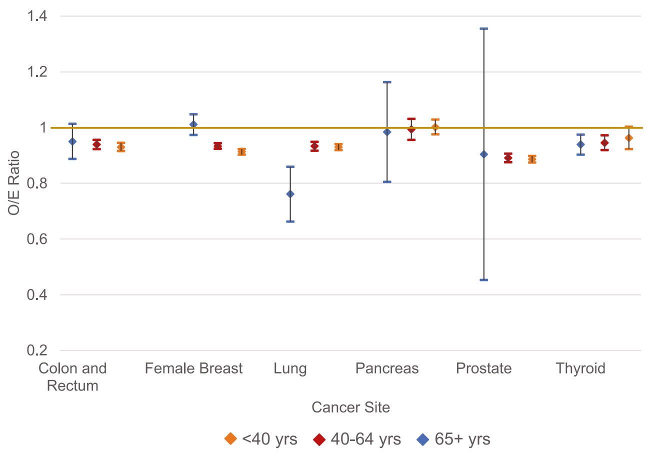

FIGURE 5 Observed-to-expected (O/E) ratios for newly diagnosed cancer cases in 2020 by age category and cancer site. The observed 2020 case count was compared with the expected count by ratio (O/E ratio). The standard error for the ratio was obtained using the delta method to determine the *p* value and 95% confidence interval for each O/E ratio. Data source for observed counts: SEER*Stat Database, North American Association of Central Cancer Registries Incidence Data—Certification year and 12-month data, 2014–2020, for the United States (which includes data from the Centers for Disease Control and Prevention's National Program of Cancer Registries, the Canadian Cancer Registry's Provincial and Territorial Registries, and the National Cancer Institute's SEER Program registries), certified by the North American Association of Central Cancer Registries as meeting high-quality incidence data standards for the specified time periods; submitted December 2021. Yrs indicates years.

impact of COVID-19 on prostate cancer counts among Black persons may reflect improved awareness of higher prostate cancer risk for Black persons or, alternatively, increased health care use during the pandemic.[33,34]

Deficit in cases diagnosed at in situ and localized stage varies by cancer site

We report statistically significantly fewer than expected cases diagnosed with localized female breast, colorectal, lung, prostate, thyroid, and pancreatic cancers. In addition, we report fewer than expected in situ tumors diagnosed for female breast, colorectal, and thyroid cancers. If indeed the underlying population with progressing but clinically undetectable tumors has not changed significantly year-to-year, it could be assumed that each case expected but not observed in 2020 is a potentially missed opportunity for early detection. For the subset of registries included in this study, 7147 colorectal cancer cases were expected to be detected at an in situ or localized stage, but only 5983 (83.7%) were diagnosed. This under-detection resulted in potentially up to 1164 (16.3%) missed opportunities for early detection. Likewise, there were 4020 fewer than expected in situ and localized female breast cancer cases (9.0%), 1267 (14.7%) fewer than expected in situ and localized lung cancer cases, and 3447 (14.8%) fewer than expected localized prostate cancer cases. Similar results were observed for the two cancer sites with no screening recommendations: thyroid (470 fewer than expected in situ and localized cases; 11.8%) and pancreas (115 fewer than expected in situ and localized cases; 9.9%). Therefore, the survival benefits of early detection may be limited for a large segment of patients with cancer.

At the same time, the number of patients diagnosed with advanced disease did not differ statistically from the expected number for female breast, thyroid, and pancreatic cancers; and the counts were marginally lower than the expected for colorectal and lung cancers. The only exception to this pattern was prostate cancer, in which the number of advanced cases was only 89% of the expected. This finding could be partially explained by the unusually high number of cases with unknown stages. To the best of our knowledge, there is no direct evidence of decreased use of positron emission tomography imaging for bone metastasis detection or of computed tomography and magnetic resonance imaging for pelvic and abdominal initial evaluation of patients with prostate cancer. However, there is indirect evidence of a more general reduction in the number of imaging procedures during the year 2020.[22] Fewer bone scans may result in an increased number of cases with incomplete risk stratification and incomplete evaluation of the extent of disease.[35]

There is a growing body of literature pointing to a decrease in screening procedures during the year 2020.[36] Data from the Behavioral Risk Factor Surveillance System indicate lower screening prevalence in 2020 than in 2012, 2014, 2016, and 2018.[37] The adjusted prevalence ratio for breast cancer screening was 0.94 (95% CI, 0.92–0.95), which is similar to the O/E ratio we report in this article for in situ tumors (i.e., 0.95), suggesting that the decline in cancer cases diagnosed in early 2020 could be explained largely by decreases in screening prevalence.[37] The results presented here are consistent with previous analyses describing the correlation of breast and colorectal cancer case counts with fewer screening events during the COVID-19 pandemic.[38,39] With cancer screening having been disrupted, detection of cancer cases may be delayed, affecting cancer survival outcomes.[40] In addition, a fraction of the overall case count decline could be caused by the excess all-cause mortality observed in 2020 in the United States.[41] Furthermore, not all states paused or reopened screening services at the same time. Thus our findings might not reflect the experience of each cancer registration jurisdiction in the United States.

Strengths and limitations

We used population-based cancer registry data, with demographic and tumor characteristics meeting NAACCR standards, from a subset of geographically diverse registries that had high 12-month data completeness. Although not necessarily representative of the US population, this is the largest study to estimate the effect of the COVID-19 pandemic on the number of newly diagnosed cancer cases in the United States.

TABLE 2 Stage-specific observed-to-expected ratios for newly diagnosed cases in 2020 by cancer type.

		Observed[a]		Expected[b]		
Cancer type	Stage	No.	%	No.	%	O/E ratio (95% CI)
Female breast	In situ	11,284	20.3	11,914	19.8	0.95 (0.93–0.97)
	Localized	29,366	52.7	32,756	54.5	0.90 (0.89–0.91)
	Advanced	13,292	23.9	13,579	22.6	0.98 (0.96–1.00)
	Unknown	1753	3.1	1815	3.0	0.97 (0.91–1.02)
Colorectal	In situ	481	2.6	645	3.2	0.75 (0.69–0.80)
	Localized	5502	30.2	6502	32.0	0.85 (0.83–0.87)
	Advanced	11,012	60.4	11,620	57.2	0.95 (0.93–0.97)
	Unknown	1231	6.8	1542	7.6	0.80 (0.75–0.84)
Lung	In situ	112	<0.1	136	<0.1	0.82 (0.60–1.04)
	Localized	7218	28.3	8461	30.4	0.85 (0.83–0.87)
	Advanced	17,396	68.1	17,992	64.7	0.97 (0.95–0.99)
	Unknown	804	3.1	1200	4.3	0.67 (0.60–0.74)
Prostate	In situ[c]	NA	NA	NA	NA	NA
	Localized	19,850	65.8	23,297	69.6	0.85 (0.84–0.86)
	Advanced	5376	17.8	6056	18.1	0.89 (0.86–0.91)
	Unknown	4950	16.4	4126	12.3	1.20 (1.16–1.24)
Thyroid	In situ	170	3.5	442	8.3	0.38 (0.31–0.46)
	Localized	3359	69.2	3557	66.8	0.94 (0.90–0.97)
	Advanced	1128	23.2	1119	21.9	1.01 (0.94–1.08)
	Unknown	196	4.0	206	3.0	0.95 (0.78–1.13)
Pancreas	In situ	30	0.5	30	0.5	1.00 (0.69–1.31)
	Localized	1014	16.7	1129	18.1	0.90 (0.81–0.99)
	Advanced	4781	78.8	4747	76.0	1.01 (0.97–1.05)
	Unknown	245	4.0	343	5.5	0.71 (0.58–0.84)

Abbreviations: CI, confidence interval; NA, not applicable/available; O/E ratio, observed-to-expected ratio.

[a]Observed counts for diagnosis year 2020 reported to the North American Association of Central Cancer Registries; 12-month submission, December 2022.

[b]Expected counts for diagnosis year 2020 based on counts reported to the North American Association of Central Cancer Registries for diagnosis years 2015–2019.

[c]Prostatic intraepithelial neoplasia (grade 3) not reportable to cancer registries.

A limitation of this study is that, given the nature of cancer reporting operations, more information about cases diagnosed in 2020 will be reported to NAACCR in subsequent years. Although we compared cases diagnosed in 2020 with comparable data sets collected for the previous 5 years, the final stage distribution of cases diagnosed in 2020 could potentially evolve with further data curation, and additional cases may be reported. The US population covered by registries for each cancer site in this analysis (18%–26%) also serves as a potential limitation to the representativeness of the data. Although overall racial minority groups were over-represented in this study (25% vs. 22% in the US population), API and AIAN populations were under-represented. In addition, persons of Asian and Pacific Islander origin are presented together in the API category. The numbers of people of Asian ancestry in the general population are much larger than the numbers of Pacific Islander persons. Other limitations are the unavailability of Hispanic ethnicity for this analysis and the lack of information about social determinants of cancer (e.g., education and socioeconomic status).

Conclusions

Our analysis provides evidence for declines in the numbers of newly diagnosed cancer cases of six major cancer types between March and May 2020, consistent with decreases in the numbers of pathology reports. This points toward declines not attributable to operational delays in cancer reporting but, instead, to missed opportunities for early detection during screening, preventive care visits, diagnostic procedures, and other health care visits. Evidence from other studies suggests that these delays in cancer detection may lead to negative long-term consequences, including shorter survival and higher mortality. Efforts to remove barriers to preventive care visits can help get people back on track for cancer screening and reduce disparities in early detection. Our findings suggest that, during the beginning of the COVID-19 pandemic in the United States, many people with cancer were not diagnosed. Novel strategies to increase guideline-concordant cancer care may help alleviate detrimental consequences on survival, mortality, and quality of life among these persons with cancer.

AUTHOR CONTRIBUTIONS

Serban Negoita: Conceptualization, methodology, formal analysis, writing–original draft, and writing–review and editing. **Huann-Sheng Chen**: Conceptualization, methodology, formal analysis, writing–original draft, writing–review and editing, and visualization. **Pamela V. Sanchez**: Conceptualization, methodology, formal analysis, writing–original draft, writing–review and editing, and visualization. **Recinda L. Sherman**: Conceptualization, methodology, and writing–review and editing. **S. Jane Henley**: Conceptualization, methodology, and writing–review and editing. **Rebecca L. Siegel**: Conceptualization, methodology, and writing–review and editing. **Hyuna Sung**: Conceptualization, methodology, and writing–review and editing. **Susan Scott**: Conceptualization and writing–review and editing. **Vicki B. Benard**: Conceptualization and writing–review and editing. **Betsy A. Kohler**: Conceptualization, writing–review and editing, and supervision. **Ahmedin Jemal**: Conceptualization, writing–review and editing, and supervision. **Kathleen A. Cronin**: Conceptualization, writing–review and editing, and supervision.

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CONFLICT OF INTEREST STATEMENT

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available directly from the North American Association of Central Cancer Registries, and release would require appropriate approvals and individual registry consent.

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